Saddle pts: $(1, 0), (-1, 4)$
Local min: $(1, 4)$

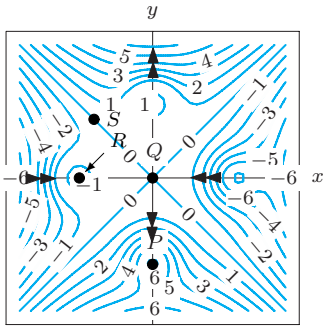
19 Saddle point: $(0, 0)$
Local max: $(1, 1), (-1, -1)$

21 $A = -2, B = 21$

23 (a) (a, b)
(b) $a = -1, b = 5$
(c) Local maximum

25 (a) Local maximum
(b) Saddle point
(c) Local minimum
(d) None of these

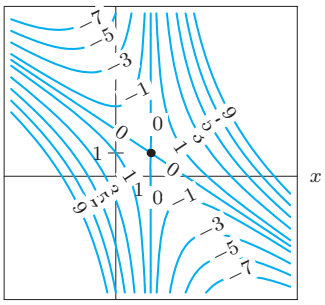
27



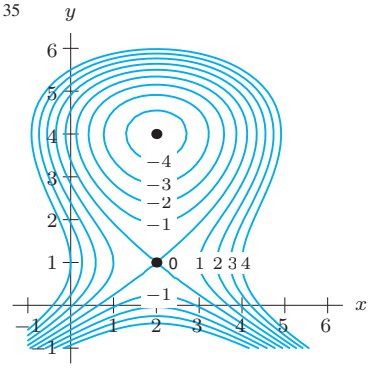
29 $(k\pi, l\pi)$,
for $k = 0, \pm1, \pm2, \cdots$,
$l = 0, \pm1, \pm2, \cdots$
$(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$,
for $k = 0, \pm1, \pm2, \cdots$,
$l = 0, \pm1, \pm2, \cdots$
$(k\pi, l\pi)$, $k = 0, \pm1, \pm2, \cdots$,
$l = 0, \pm1, \pm2, \cdots$ are saddle points
If k and l are both even or k and l are both odd, then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local max
If k is even but l is odd or k is odd but l is even, then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local min

31 $y = 0, \pm2\pi, \pm4\pi, \ldots$ Local minima

33 (a) (a, b) is a saddle point.
(b)

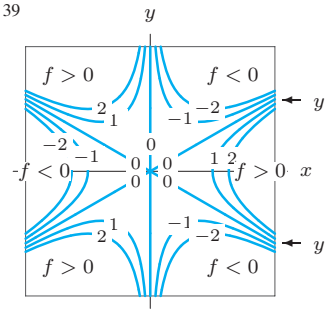


35



37 (a) $(0, 0)$
(b) $D = -24x^2$
(c) Saddle point

39



41 $(1, 3)$ could be saddle point

43 Can be saddle if f_{xy} large

45 $f(x, y) = 4 - (x - 2)^2 - (y + 3)^2$

47 False

49 True

51 False

53 True

55 False

Section 15.2

1 Mississippi:
$87 - 88$ (max), $83 - 87$ (min)
Alabama:
$88 - 89$ (max), $83 - 87$ (min)
Pennsylvania:
$89 - 90$ (max), 80 (min)
New York:
$81 - 84$ (max), $74 - 76$ (min)
California:
$100 - 101$ (max), $65 - 68$ (min)
Arizona:
$102 - 107$ (max), $85 - 87$ (min)
Massachusetts:
$81 - 84$ (max), 70 (min)

3 Max: 30.5 at $(0, 0)$
Min: 20.5 at $(2.5, 5)$

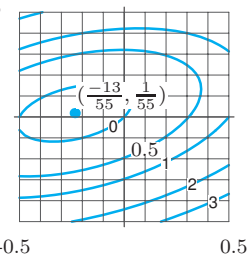
5 Min $= 0$ at $(0, 0)$
(not on boundary)
Max $= 2$ at $(1, 1), (1, -1)$, $(-1, -1)$ and $(-1, 1)$
(on boundary)

7 max$= 1$ at $(1, 0)$ and $(-1, 0)$
(on boundary)
min$= -1$ at $(0, 1), (0, -1)$
(on boundary)

9 Global min

11 Global max $= 0$
No global min

13 (a) Local min: $(-13/55, 1/55)$
(b)



15 All edges $(32)^{1/3}$ cm

17 $l = w = h = 45$ cm

19 $(3/14, 1/7, 1/14)$

23 $q_1 = 300, q_2 = 225.$

25 (a) $L = \left[pA\left(\frac{a}{k}\right)^a \left(\frac{l}{b}\right)^{(a-1)}\right]^{1/(1-a-b)}$
$K = \frac{la}{kb}L$
(b) No

27 $y = 24x^2/49 - 2/7$

31 (a) $p = \sqrt{P_0 P_F}$
(b) $p_1 = \sqrt[3]{P_0^2 P_F}, p_2 = \sqrt[3]{P_0 P_F^2}$

33 Must have global maximum if continuous and R closed, bounded

35 Local max not necessarily global max

37 $f(x, y) = x^2 + y^2$, R is $0 \le x, y \le 1$

39 False

41 False

43 False

45 True

Section 15.3

1 Min $= -\sqrt{2}$, max $= \sqrt{2}$

3 Max: 20 at $(-1, 2)$;
Min: 0 at $(1, -2)$

5 Min $= -22$, max $= 22$

7 Global min: $1/2$
No global max

9 Min $= -\sqrt{35}$, max $= \sqrt{35}$

11 Max: 4 at $(2, 2, 1), (2, -2, -1)$, $(-2, 2, -1), (-2, -2, 1)$;
Min: -4 at $(-2, -2, -1), (-2, 2, 1)$, $(2, -2, 1), (2, 2, -1)$

13 Max $= f(\frac{1}{\sqrt{5}}, \frac{3}{\sqrt{5}}) = 2\sqrt{5}$
Min $= f(-\frac{1}{\sqrt{5}}, -\frac{3}{\sqrt{5}}) = -2\sqrt{5}$

15 No global extrema

17 Max: $30 + \sqrt{10}$ at $(0, -\sqrt{10})$;
Min: -5 at $(\pm3, -1)$

19 1 at $(1/2, 1/2)$

21 0.5